Cross sections of solids

1

a Yes

b Triangular Prism

2

a A hexagon

b Yes

c A hexagonal prism

3 **C**

4 **D**

5 **D**

6

a Square

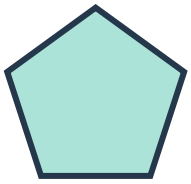
b Circle

c Rectangle

d Triangle

7

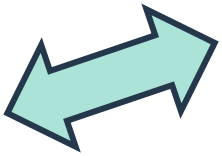
a



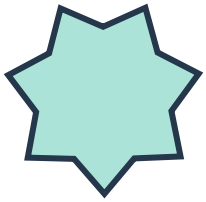
b



c



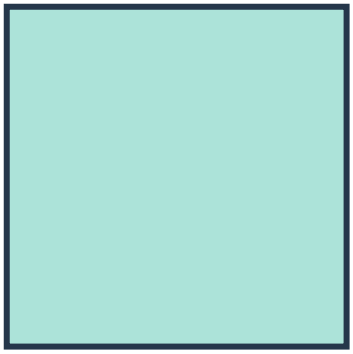
d



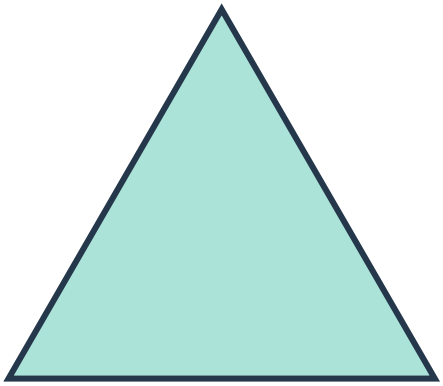
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8

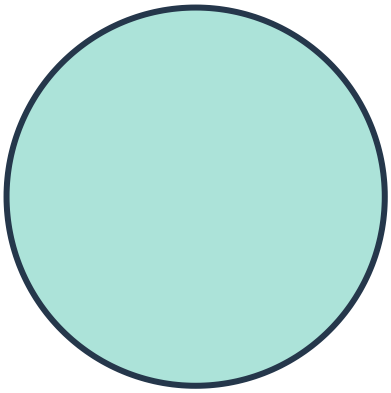
a



b



c



d

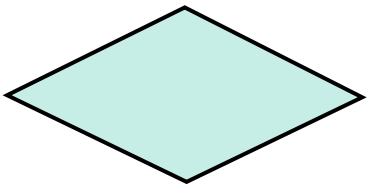


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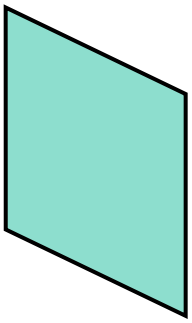
9

a

i

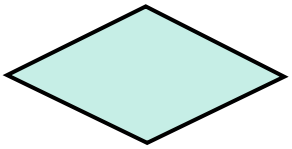


ii

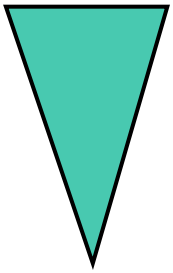


b

i



ii



Solids from cross sections

10

a No **b** Yes **c** Yes **d** No

11

a Yes **b** Yes **c** No **d** No

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12 Example answers:

- Rectangular Pyramid
- Rectangular prism
- Triangular Prism

13 Example answers:

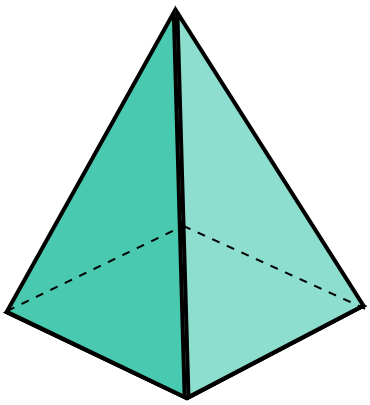
- Pentagonal Pyramid
- Pentagonal Prism

14

a

i A square pyramid, or trapezoidal prism

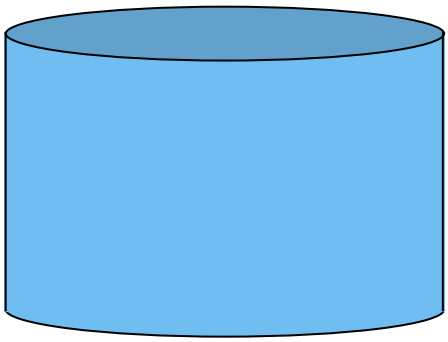
ii



b

i A sphere, cone, or cylinder

ii



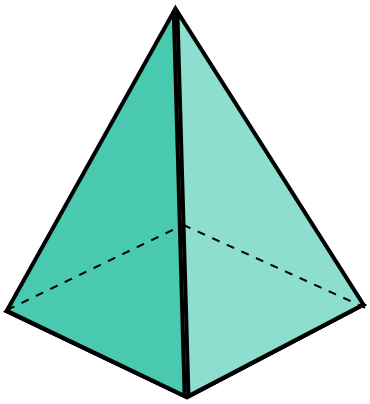
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Applications

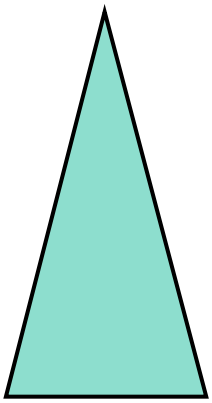
15 Hiroki is correct about the sphere. Any cross section of a sphere will produce a circle. There are not any other solids where this property holds true.

16 No. As the perpendicular cross section can be taken anywhere perpendicular the base, if the shape is a pyramid then a perpendicular cross section may be smaller than the two dimensional side view.

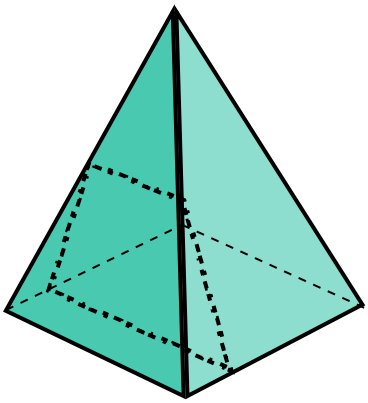
A square-based pyramid



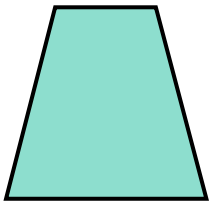
Side view



Perpendicular cross section location



Cross section



17 It is possible to take a cross section parallel to the epidermis and see all of the components shown in the perpendicular cross section. However, it is possible to take a parallel cross section of the root only see some of the parts of the root. Viewing the perpendicular cross section would guarantee you can examine all cellular components of the root.

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